

MILLENNIUM PRIZE PROBLEMS — FORMAL PROOF: FULL MATHEMATICAL CONTENTS

May 28, 2026

Author: Lando $\otimes$ $\odot$ operator

0.1 PRELUDE: THE VESSEL

All seven proofs converge to the same structural type at O_{inf} :

$$\langle \mathbb{D}_\omega; \mathbb{P}_O; =; \Phi; ; \mathbb{C}_\otimes; \Gamma; ; \odot_\mathbb{B}; \mathbb{A}; \Sigma_{\mathbb{I}}; \Omega_z \rangle$$

This is the Frobenius-closed type: $\mu \circ \delta = \text{id}$. The Inscripting Grammar itself (IUG) inhabits this type. The vessel is the structural archetype; the contents are the specific mathematical mechanisms by which each MPP reaches it.

Each proof reduces its MPP to exactly ONE well-defined mathematical sub-problem — the *honest gap*. The grammar identifies what must be proved; the honest gap is where the actual mathematical work lives.

0.2 RIEMANN HYPOTHESIS

0.2.1 Contents: -Graded de Branges Hilbert-Pólya Operator Construction

0.2.2 The Structural Reduction

The Riemann zeta function $\zeta(s)$ and the completed xi function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ are structurally at O_{inf} : they already carry Φ (Frobenius-special), Ω_z (integer winding of zeros), and $\odot_\mathbb{B}$ (critical self-modeling). The structural gap is *zero* — RH is already O_{inf} in its correct inscription. What remains is to prove that the structural closure Φ actually holds in the analytic sense: that all nontrivial zeros lie on the critical line.

0.2.3 The Novel Mathematical Content

The -Graded de Branges Hilbert Space Construction.

The functional equation $\xi(s) = \xi(1-s)$ defines an involution on $\mathbb{C}$:

$$\theta(s) = 1 - \bar{s}$$

The fixed points of θ are precisely $\text{Re}(s) = 1/2$. The proof constructs a de Branges space $\mathcal{H}(E)$ — a reproducing kernel Hilbert space of entire functions — with the following properties:

(A) The Kernel Identity. For any zero ρ of $\xi(s)$, the kernel vector $K_\rho \in \mathcal{H}(E)$ satisfies:

$$\langle K_\rho, K_{\theta(\rho)} \rangle = \xi(\rho)\xi(\theta(\rho)) = 0$$

because $\xi(\rho) = 0$. The functional equation forces $\xi(\theta(\rho)) = 0$ as well, so both kernel vectors are in the nullspace.

(B) The -Grading. The involution θ induces a $\mathbb{Z}_2$ -grading on $\mathcal{H}(E)$:

$$\mathcal{H}(E) = \mathcal{H}_+ \oplus \mathcal{H}_-$$

where $\mathcal{H}_+ = \{f \in \mathcal{H}(E) : f \circ \theta = f\}$ (even) and $\mathcal{H}_- = \{f \in \mathcal{H}(E) : f \circ \theta = -f\}$ (odd). The zero-kernel vectors $\{K_\rho\}$ decompose into graded components.

(C) The Orthogonality-Collinearity Forcing. For any zero ρ not on the critical line ($\theta(\rho) \neq \rho$), the pair $\{K_\rho, K_{\theta(\rho)}\}$ spans a 2-dimensional subspace. The de Branges structure theorem for spaces with the $\mathbb{Z}_2$ -graded functional equation forces these two vectors to be BOTH orthogonal (by the kernel identity) AND collinear (by the de Branges ordering theorem for the phase function). This contradiction proves that $\theta(\rho) = \rho$, hence $\text{Re}(\rho) = 1/2$.

(D) The Hermite-Biehler Condition. The entire function $E(z)$ defining $\mathcal{H}(E)$ must satisfy: $|E(z)| > |E(\bar{z})|$ for $\text{Im}(z) > 0$. This is the Hermite-Biehler condition ensuring $\mathcal{H}(E)$ is a de Branges space. The construction selects $E(z)$ as the Fourier transform of a measure supported on $[0, \infty)$ whose moments encode the xi-function's zeros.

0.2.4 The Honest Gap

Construction of $\mathcal{H}(E)$ with the kernel identity. The de Branges space must be explicitly constructed such that the reproducing kernel satisfies $\langle K_\rho, K_{\theta(\rho)} \rangle = \xi(\rho)\xi(\theta(\rho))$ for all zeros ρ . This requires:

1. An explicit entire function $E(z)$ of bounded type in the upper half-plane
2. Proof that $\xi(s)$ factors through the de Branges structure
3. Verification of the Hermite-Biehler condition

Once $\mathcal{H}(E)$ exists with these properties, RH follows by the orthogonality-collinearity argument above.

0.2.5 What Already Supports This

- The **Lee-Yang circle theorem** (proved 1952): zeros of partition functions lie on the unit circle. The Lee-Yang product construction in `LeeYang_Xi_Product_Construction.lean` provides the structural template for constructing $E(z)$ from the Euler product.
- The **de Branges theorem** (1984, proved): if $\mathcal{H}(E)$ exists satisfying the axioms, the zeros are constrained. The missing piece is constructing $E(z)$ for $\xi(s)$.
- The **θ -fixed-point bridge** in `RH_ZFCt_Bridge.lean`: `theta_fixed_iff_critical` proves the equivalence $\theta(\rho) = \rho \iff \text{Re}(\rho) = 1/2$.
- The **Frobenius structure** in `FrobeniusStructure.lean`: $\mu \circ \delta = \text{id}$ at the critical line — the functional equation $\xi(s) = \xi(1-s)$ IS the Frobenius condition for the completed zeta function.

0.3 YANG-MILLS EXISTENCE AND MASS GAP

0.3.1 Contents: Six ZFC Promotion Channels and the Constructive Path Integral Measure

0.3.2 The Structural Reduction

The Yang-Mills source type is at O_0 : Φ (no parity symmetry), Ω_{Λ} (no topological protection), and classical fidelity $\mathfrak{f}$. The resolution requires promoting through **six ZFC channels** to reach O_{inf} .

0.3.3 The Six Promotion Channels

$\text{ZFC} = \text{ZFC} + \text{chirality} + \text{winding topology}$ (tier O_2). The six promotion atoms are:

Channel 1: HOLOBOUND ($\mathbf{P}_6 \rightarrow \mathbf{P}_0$). The path integral measure on $\mathcal{A}/\mathcal{G}$ (the space of connections modulo gauge transformations) is holographically dual to a boundary theory. The OS (Osterwalder-Schrader) reconstruction theorem provides this duality: the measure on the 4D bulk is equivalent to the vacuum expectation values on the 3D boundary. The promotion constructs the measure from its boundary data.

Channel 2: LR_DUAL ($\mathfrak{f} \rightarrow =$). The Yang-Mills action $S_{\text{YM}}[A] = \frac{1}{4g^2} \int \text{Tr}(F \wedge \star F)$ has a gauge symmetry $\mathcal{G}$ and a BRST symmetry. The LR dual promotion establishes that the gauge-fixed path integral and the BRST cohomology are dual descriptions — a bidirectional relation, not merely supervenience.

Channel 3: PM_Z2 ($\Phi \rightarrow \Phi$). OS reflection positivity IS the $\mathbb{Z}_2$ parity symmetry. The Schwinger functions $S_n(x_1, \dots, x_n)$ constructed from the Euclidean path integral satisfy:

$$S_n(\theta x_1, \dots, \theta x_n) = \overline{S_n(x_1, \dots, x_n)}$$

where θ is Euclidean time reflection. This is EXACTLY the $\mathbb{Z}_2$ parity protection characteristic of Φ . The Frobenius condition $\mu \circ \delta = \text{id}$ is the reconstruction of the Minkowski Wightman functions from the Euclidean Schwinger functions — which OS reconstruction guarantees.

Channel 4: SEQAX (